

Two-Stage Fusion for Routing and Classification in Parameter-Efficient Continual Learning

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Abstract. Task-specific parameter-efficient continual learners maintain adapters whose selection affects both routing and classification. We augment HRM-PET with a random-projection (RP) head fitted by ridge regression over fixed first-task LoRA features. Its sufficient statistics are accumulated task by task without retaining examples. The same RP scores support two inference-time operations: task-level fusion with the initial adapted prediction before re-matching, and class-level fusion with the final logits using a top-two-margin gate. Across three benchmarks and four seeds, DRM-proposal accuracy increases by 1.10–2.26 points and classification accuracy by 0.69–1.50, with no consistent improvement in forgetting or backward transfer. Single-seed ablations across five backbones show that the routing contribution depends on the gate. Four-seed conditional contrasts show a positive routing contribution with gated class fusion on ImageNet-R and CUB-200, but an unresolved effect on CIFAR-100. The gate’s estimated contribution is negative on CUB-200, with a confidence interval including zero. A grid over the two mixing weights shows that classification accuracy is nearly flat in the routing weight, so the reported value is not a resolved optimum.

Keywords: Continual learning · Parameter-efficient tuning · Parameter routing · Random projection · Signal fusion.

1 Introduction

Continual learning asks a model to absorb a stream of tasks without losing what earlier ones taught it. Parameter-efficient tuning adapts a frozen pre-trained backbone using a small number of parameters. One important family maintains task-specific prompts or adapters and selects them at inference [5, 3]. Such methods can avoid storing old examples, but preserving parameters does not guarantee preserved accuracy.

In this family, task identity is unknown at inference and parameter selection is itself a prediction. A wrong adapter can produce unsuitable features. Larger pools, attention-weighted mixtures and explicit re-matching offer different responses to this difficulty [4, 7, 1]. We focus on HRM-PET, where routing and classification use related predictions from the adapted and unadapted backbone.

We add a complementary predictor: first-task adapter features, a frozen random projection and a ridge classifier fitted from accumulated statistics, following RanPAC [2]. Fitting uses current-task data without changing LoRA/TII training or retaining examples. Its scores revise the task proposal before re-matching and blend with final class logits, with greater weight when the primary classifier’s top-two margin narrows.

Contributions. We identify complementary errors in HRM-PET (Sect. 2.1) and introduce two inference-time fusion operations using one RP head: task-proposal fusion before re-matching and gated class fusion afterwards. The head is fitted in closed form without gradient descent. A three-dataset, four-seed study separates gains in routing/classification from unresolved retention effects. Conditional ablations and weight comparisons examine what each operation adds. We also report a historical four-dataset by five-pretraining grid, with its verification limits stated in Sect. 4.

2 Related Work

Parameter pools and the selection they require. L2P [4] and Dual-Prompt [5] exemplify pool-based parameter-efficient continual learning: freeze the backbone, keep a pool of lightweight parameter sets, and pick from it at inference using a query from the bare backbone. CODA-Prompt [7] replaces the discrete pick with an attention-weighted combination; HiDe-Prompt [3] separates within-task prediction, task-identity inference and task-adaptive prediction and optimises them apart, observing that identity inference, not within-task classification, is what limits the pipeline. HRM-PET [1], our baseline, substitutes LoRA adapters [6] for prompts and adds re-matching stages that revisit the initial decision. Later work enlarges the expert family: MoTE [14] combines task-specific experts, and DLEPEM [15] combines general and adapted prototypes for matching.

Closed-form solutions in continual learning. Closed-form classifiers preserve past contributions through additive statistics of fixed features. APER [8] adapts the backbone on the first task, then uses frozen features and class prototypes. ACIL [20] uses recursive least squares, replacing a sample buffer with correlation statistics and recovering the corresponding joint solution. DS-AL [21] adds a second analytic stream. RanPAC [2] applies a fixed random projection to frozen features, accumulates second-order statistics and solves a ridge system.

Our head belongs to this family and we claim no novelty in it: the projection, the Gram accumulation and the ridge solve are RanPAC’s. What differs is its role. These methods use the closed-form classifier as the whole predictor and discard routing; we keep the routing pipeline and consult the solved head twice.

Correcting the classifier after the fact. Class-incremental learning often biases predictions towards recent classes [9]. iCaRL [10] uses nearest exemplar means; BiC [11] learns an affine logit correction on held-out data; Weight Aligning [12] matches new- and old-class weight norms; and LUCIR [13] normalises features and weights. Our fusion leaves the adapters, auxiliary classifier and shared head fixed. It retains no exemplars but requires current-task data to fit the RP head.

2.1 HRM-PET: The Baseline We Build On

We use the released HRM-PET [1] inference path. Diagnostics quoted here are seed-42 measurements, separate from the evaluation of Sect. 4.

Task-identity inference (TII). The input image is passed through the *bare* backbone, with no adapter attached, to obtain a query feature $q(x) = \phi(x; \theta_{\text{ptm}})$, writing ϕ for the backbone under whatever adapter it is given. That feature enters an auxiliary classifier trained with cross-entropy, which gives unnormalised class logits $u = g(q(x); \theta_\omega)$. The initial match is then $t_0 = \mathcal{T}(\arg \max_c u_c)$.

Re-matching (DRM and CRM). The first adapted pass attaches p_{t_0} and produces $r = g_{\theta_g}(\phi(x; p_{t_0}, \theta_{\text{ptm}}))$. DRM proposes the task of $\arg \max r$ and evaluates its adapter when it differs from t_0 ; our fusion modifies this proposal, not the preceding bare TII pass. CRM selects low-confidence samples from r by the baseline’s generalised-entropy threshold and compares the current logits, under the baseline log-sum-exp score, with the adapter of the second-ranked class of r ; it is confidence-triggered, not restricted to cases where the two task identities agree. Both reuse trained modules, can require further adapter forwards, and are heuristics: neither guarantees that accuracy cannot decrease.

Shared classification head. The selected parameter set is attached to the backbone, and the shared head produces a score vector over all classes seen so far, $\ell = g(\phi(x; p_{\hat{t}}, \theta_{\text{ptm}}); \theta_g) \in \mathbb{R}^{|\mathcal{Y}|}$, predicting $\hat{y} = \arg \max_c \ell_c$, where $\hat{t}$ is the identity after DRM and CRM. Classes not yet seen are masked with $-\infty$.

Two weaknesses this paper acts on. The baseline draws its routing evidence from TII and the shared head alone. To see what a second source would be worth we built the head of Sect. 3.2 on HRM-PET’s own LoRA features and used it as a router: at seed 42 on ImageNet-R, TII routes 63.80% of samples correctly, the DRM proposal 77.79% and the RP head 77.12%, while the union of TII and the head reaches 82.99%. Since TII is wrong on 36.20% of the set, the head repairs 53.0% of its errors, against 62.0% and 46.9% on CIFAR-100 and CUB-200. The union is an oracle that no router we build attains, and it compares bare TII with RP rather than the actual Stage 1 inputs.

A better route need not become a better prediction: ℓ comes from the shared head, which spans every class seen so far, so a mis-routed sample is often classified correctly anyway. The RP head is moreover a complete classifier that a routing-only variant would discard: in the historical routing-only diagnostic ($w = 0.7$, $\beta = 0$) on ImageNet-R, the routed head reaches 74.31% and the projection head 70.08%, yet their union is 78.89%, the head alone contributing 4.58 points of the test set and 2.43 and 3.87 on the other two. These measurements give the two components described next.

3 Method

3.1 Problem Statement

Class-incremental learning (CIL) learns a sequence of T tasks without forgetting earlier ones. The t -th task consists of n_t samples $\mathcal{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ drawn from a class set $\mathcal{C}_t$, where $x_i^t \in \mathcal{X}$ is an input image and $y_i^t \in \mathcal{Y}$ its label.

Three constraints define the setting. At step t only $\mathcal{D}_t$ is available for training. The class sets are disjoint, $\mathcal{C}_t \cap \mathcal{C}_{t'} = \emptyset$ for all $t \neq t'$, and at inference the *task identity is unknown*: the model must discriminate among all classes seen so far through a single shared classification head. Finally, this work operates in the exemplar-free setting: images from past tasks are never retained.

In the task-specific PET setting considered here [3, 1], a pre-trained backbone h_{ptm} with parameters θ_{ptm} is frozen and a lightweight parameter pool $\mathcal{P} = \{p_1, \dots, p_T\}$ is maintained, each set trained on one task; here every p_t is a LoRA adapter [6]. The shared classification head is denoted g_{θ_g} . Because the task identity is unknown at inference, the system is forced to *choose* which parameter set to attach for its primary prediction. Our fixed RP branch does not depend on this inferred task identity.

We distinguish four class-score vectors: bare TII scores u , initial adapted shared-head logits r , final re-matched logits ℓ , and RP scores s . The class-to-task map is $\mathcal{T}(\cdot)$; t_0 is the initial TII match and $\hat{t}_{\text{DRM}}$ the task proposed for direct re-matching. We write $z(\cdot)$ for per-sample standardisation with the population standard deviation floored at 10^{-6} .

3.2 The Second Source of Evidence

We fit an RP head after each task and reuse its scores in two inference-time fusion operations. HRM-PET parameters are not retrained. Figure 1 distinguishes the initial adapted logits from the bare TII scores and shows both reuse paths.

The RP head follows RanPAC [2]. Features $f \in \mathbb{R}^d$, $d = 768$, are column vectors from ViT-B/16 with the first task’s LoRA adapter (p_1 here, index 0 in code), fixed during accumulation and inference and unnormalised; changing adapters between tasks would mix incompatible spaces.

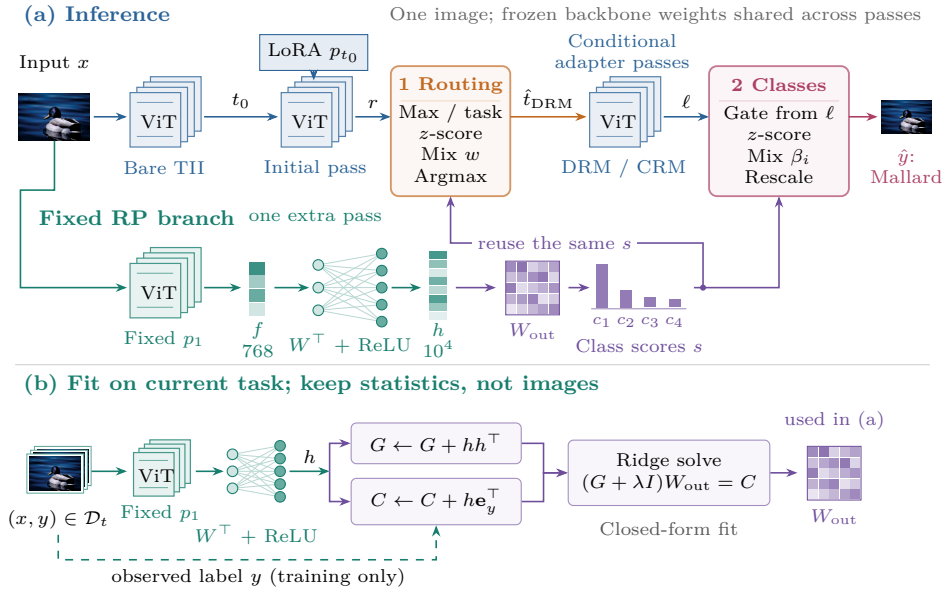


Fig. 1. Two-stage fusion. All inference passes use the same image; adapted passes share head g . TII selects t_0 ; the projection W is frozen. Stage 1 standardises task-wise maxima of r and s ; Stage 2 uses the top-two probability margin of ℓ for its continuous gate (Eqs. 6–8), then predicts by argmax. DRM runs if $\hat{t}_{\text{DRM}} \neq t_0$; CRM uses low-confidence r and its second-ranked class’s task, comparing logits by log-sum-exp (not at task 1). Only seen classes/tasks participate; score bars are schematic. Photo: T. G. Barnes/USFWS (public domain).

Projection and second-order statistics. The feature is projected into a high-dimensional space of size $M = 10^4$ and passed through a nonlinearity, and the projected features are accumulated into two matrices:

$$h = \text{ReLU}(W^\top f) \in \mathbb{R}^M, \quad W \in \mathbb{R}^{d \times M}, \quad W_{jk} \sim \mathcal{N}(0, 1), \quad (1)$$

$$G = \sum_n h_n h_n^\top \in \mathbb{R}^{M \times M}, \quad C = \sum_n h_n \mathbf{e}_{y_n}^\top \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad (2)$$

where $\mathbf{e}_y$ is the basis vector of class y . Each current-task training sample contributes once; G and C are accumulated in float64. W is generated once with seed 1993 and then frozen.

The nonlinear expansion permits richer decision functions than a linear head on f . Without ReLU, $W^\top f$ remains linear in f and adds no separating power. This does not guarantee that the expanded data become linearly separable.

Classification by ridge regression. The output weights are the closed-form solution of a regularised least-squares problem:

$$W_{\text{out}} = (G + \lambda I)^{-1} C \in \mathbb{R}^{M \times |\mathcal{Y}|}, \quad s = W_{\text{out}}^\top h \in \mathbb{R}^{|\mathcal{Y}|}. \quad (3)$$

The implementation solves the regularised system rather than inverting it.

Two inherited properties. Both G and C in (2) are *sums* over samples, so for a *fixed* feature map they, and with them W_{out} , do not depend on the order in which samples arrived: nothing an earlier task contributed is overwritten, and the head after T tasks equals the ridge solution of a joint fit over every feature seen. Their sizes do not grow with the sample count, and no per-sample feature is retained. This exact-arithmetic identity does not imply bitwise order invariance in floating point or an absence of forgetting: new classes can outscore old ones despite unchanged statistics. Moreover, changing the first task changes its adapter and hence the feature map.

3.3 Two-Stage Fusion

Both stages blend sources with different score scales. Direct mixing can make the nominal weights poorly reflect their influence. Each source is standardised per sample:

$$z(v) = \frac{v - \text{mean}(v)}{\sigma_\epsilon(v)}, \quad \sigma_\epsilon(v) = \max\{\text{std}(v), 10^{-6}\}, \quad (4)$$

where std is the population standard deviation. Moments are over seen tasks in Stage 1 and seen classes in Stage 2, excluding unseen entries.

First stage: fusion at the routing level. Stage 1 runs after the initial adapted pass and before DRM. Its input is r , the shared-head logits under the adapter selected by bare TII, not auxiliary logits u . Both r and RP scores s are mapped to seen tasks by $r_t^{\text{task}} = \max_{c \in \mathcal{C}_t} r_c$ and $s_t^{\text{task}} = \max_{c \in \mathcal{C}_t} s_c$ for $t = 1, \dots, \hat{T}$, with $\hat{T}$ the number of tasks seen so far; classes of unseen tasks never enter either maximum. The two vectors are blended after standardisation:

$$\text{fused} = w z(r^{\text{task}}) + (1 - w) z(s^{\text{task}}), \quad \hat{t}_{\text{DRM}} = \arg \max_t \text{fused}_t, \quad w = 0.7. \quad (5)$$

For one seen task the standard deviation is zero, so the floor keeps the operation finite. The fused task replaces the baseline DRM proposal $\mathcal{T}(\arg \max r)$; the initial TII match, CRM candidate rule, confidence test and shared head are unchanged. At $w = 1$ the ordering is that of r^{task} , recovering that proposal; the full baseline also needs $\beta = 0$.

Second stage: gated fusion at the classification level. Because ℓ comes from the shared head, it frequently predicts the correct class in spite of a wrong routing decision, so we use the projection head itself as a secondary classifier, this time taking its class scores s . A fixed blend would alter confident primary predictions; we instead reduce the RP share as the top-two probability margin grows, while recognising that confident predictions can still be wrong:

$$\beta_i = \beta \left(1 - \frac{p_{(1)} - p_{(2)}}{p_{(1)} + p_{(2)}} \right), \quad \beta = 0.5, \quad (6)$$

where $p_{(1)} \geq p_{(2)}$ are the two largest probabilities of $\text{softmax}(\ell)$ over the classes seen so far. Dividing by the sum makes the quantity depend only on the ratio $p_{(2)}/p_{(1)}$, so the gate measures how threatened the top class is rather than how much mass the pair holds. The RP share falls to zero as the top class dominates and reaches β when the two are equal: a continuous weight, not a binary test.

The blend reuses (4) and then maps the result back onto the scale of ℓ :

$$m_i = (1 - \beta_i) z(\ell_i) + \beta_i z(s_i), \quad (7)$$

$$\hat{\ell}_i = m_i \cdot \sigma_\epsilon(\ell_i) + \text{mean}(\ell_i) \in \mathbb{R}^{|\mathcal{Y}|}. \quad (8)$$

The positive affine map-back preserves rankings and restores a reference scale for cross-entropy, not a calibration guarantee: for standardised nondegenerate vectors with correlation ρ and the floor inactive, the mixture variance is $(1 - \beta_i)^2 + \beta_i^2 + 2\beta_i(1 - \beta_i)\rho$, which need not equal one. At $\beta = 0$ this stage leaves ℓ unchanged.

Cost. At inference the head costs one extra adapter forward per sample, serving both stages. That is the head’s own cost, not the pipeline’s difference, which we did not measure: changing the DRM proposal can change the number of adapter evaluations.

Per task the head accumulates G and C over the current task’s training split and solves one ridge system, so its footprint is set by M , not by how many samples have been seen. At $M = 10^4$ the Gram matrix holds 10^8 float64 entries, or 800 MB. Together with its regularised copy and Cholesky factor, these three matrices require approximately 2.4 GB; this is an analytical estimate, not a measured total-memory peak. A historical ten-task evaluation on ImageNet-R took 283 s on one RTX 4090 at batch 24; neither its phase-level costs nor its overhead over the baseline were measured.

4 Experiments

4.1 Setup

The backbone is ViT-B/16 [22], pre-trained on ImageNet-21K. Rank-8 LoRA modifies the key/value projections in the first five blocks. The four-seed study splits ImageNet-R [23], CIFAR-100 [24] and CUB-200-2011 [25] into 10 tasks of 20, 10 and 20 classes, respectively. The grid replaces CUB with ImageNet-A [26] (10×20 classes) and the five-dataset protocol of [1] (one domain per task). Its pre-training settings are Sup-21K, MoCo-v3 [27], iBOT-1K/iBOT-21K [28] and DINO [29].

Each seed (42–45) has separately trained LoRA/TII parameters; pre-trained backbone weights remain frozen. Both methods use identical saved checkpoints and evaluation paths. RP fits current-task training images once using the evaluation transform: resize to 256, centre crop to 224, and convert to a tensor; CIFAR additionally uses its configured channel normalisation. No test labels enter the fit. We use $M = 10^4$, ReLU, $\lambda = 10^4$, no additional RP feature normalisation or temperature calibration, and no fusion ramp. Fusion starts at task 1; CRM retains **En=gen**, threshold -10 , and the second-ranked class candidate. Training/evaluation scripts and dataset preprocessing are versioned in the code repository [30].

Acc@task in the released evaluator measures the DRM task proposal. In the default class-based routing path, CRM can change the output logits without updating the task identity passed to this metric; it is therefore not a measure of final adapter selection after CRM. Acc@1, Acc@5 and Loss use the final classification logits. Final metrics average the per-task values with equal task weights, not by pooling all test images. Writing $a_{i,t}$ for classification accuracy on task i at stage t , the retention measures are

$$\text{Forgetting} = \frac{1}{T-1} \sum_{i=1}^{T-1} \left(\max_{i \leq t \leq T} a_{i,t} - a_{i,T} \right), \quad (9)$$

$$\text{Backward} = \frac{1}{T-1} \sum_{i=1}^{T-1} (a_{i,T} - a_{i,i}). \quad (10)$$

Both averages exclude the last task, and the maximum includes only stages at which task i has been learned. Lower is better for Loss and Forgetting.

Sanity check. $w = 1$ restores the baseline DRM proposal and $\beta = 0$ bypasses class fusion. After correcting unseen-class masking, this identity matches the baseline to logged precision at every stage for ImageNet-R/CIFAR seeds 42–45 and CUB seed 42. For example, ImageNet-R seed 42 gives Acc@task 77.7914, Acc@1 74.0191 and Loss 1.2305. CUB seeds 43–45 and the historical grid/ablations still require complete post-fix verification; identity agreement alone does not validate them.

4.2 Main Results

Table 1 reports all three datasets over four seeds. The change column is pairwise — taken per seed, then averaged — so its standard deviation describes the stability of the improvement, not the seeds’ difficulty.

Table 1. Final results, mean $\pm$ sample SD over seeds 42–45; Change is paired full minus baseline. Accuracy/retention are percentages (changes in points). CUB values are historical pending complete post-fix verification. *Pending* denotes unverified corrected absolute statistics.

Dataset	Metric	HRM-PET	Proposed	Change
<i>ImageNet-R</i>	Acc@task	77.58 $\pm$ 0.33	79.85 $\pm$ 0.18	+2.264 $\pm$ 0.158
	Acc@1	73.94 $\pm$ 0.48	75.18 $\pm$ 0.24	+1.236 $\pm$ 0.416
	Acc@5	86.71 $\pm$ 0.21	88.02 $\pm$ 0.17	+1.312 $\pm$ 0.148
	Loss	1.23 $\pm$ 0.01	1.19 $\pm$ 0.01	−0.040 $\pm$ 0.002
	Forgetting	3.36 $\pm$ 0.56	3.30 $\pm$ 0.35	−0.056 $\pm$ 0.334
	Backward	−3.16 $\pm$ 0.61	−3.20 $\pm$ 0.42	−0.047 $\pm$ 0.282
<i>CIFAR-100</i>	Acc@task	89.77 $\pm$ 0.11	90.87 $\pm$ 0.09	+1.103 $\pm$ 0.066
	Acc@1	89.78 $\pm$ 0.03	90.47 $\pm$ 0.05	+0.690 $\pm$ 0.063
	Acc@5	98.65 $\pm$ 0.03	98.86 $\pm$ 0.07	+0.218 $\pm$ 0.057
	Loss	0.39 $\pm$ 0.00	0.39 $\pm$ 0.00	−0.0003 $\pm$ 0.0012
	Forgetting	3.89 $\pm$ 0.08	<i>Pending</i>	−0.150 $\pm$ 0.095
	Backward	−3.87 $\pm$ 0.07	<i>Pending</i>	+0.136 $\pm$ 0.095
<i>CUB-200</i>	Acc@task	93.12 $\pm$ 0.14	94.34 $\pm$ 0.10	+1.226 $\pm$ 0.149
	Acc@1	86.38 $\pm$ 0.25	87.88 $\pm$ 0.27	+1.495 $\pm$ 0.353
	Acc@5	97.04 $\pm$ 0.10	97.44 $\pm$ 0.13	+0.400 $\pm$ 0.103
	Loss	0.57 $\pm$ 0.00	0.53 $\pm$ 0.01	−0.036 $\pm$ 0.004
	Forgetting	2.54 $\pm$ 0.35	2.18 $\pm$ 0.12	−0.359 $\pm$ 0.325
	Backward	−2.42 $\pm$ 0.39	−2.14 $\pm$ 0.13	+0.279 $\pm$ 0.391

Intervals are paired 95% Student- t , $\bar{\Delta} \pm 3.182 s_{\Delta} / \sqrt{4}$: imprecise at four seeds, and assuming approximately normal seed-wise differences. Correlated metrics mean we discuss axes rather than count confirmations.

Routing (Acc@task): improved. The gain is +1.10 to +2.26 points, with positive paired intervals on all three datasets. This is the axis the first stage acts on directly.

Classification (Acc@1, Acc@5): improved. Acc@1 rises by +0.69 to +1.50 points and Acc@5 by +0.22 to +1.31, with positive paired intervals for both metrics on all three datasets. The full method’s Acc@1 improvement repeats across every seed. These aggregate gains do not establish how many routing repairs also repair classification.

Retention (Forgetting, Backward): unresolved. All six paired retention intervals include zero. Corrected CIFAR Forgetting is -0.150 $[-0.302, +0.002]$ and Backward $+0.136$ $[-0.015, +0.287]$. These related metrics are not independent confirmations. Neither consistent improvement nor equivalence is established: four seeds leave substantial uncertainty, and no equivalence margin was specified. The historical single-seed decomposition in Sect. 4.4 is descriptive.

4.3 Placement Against Prior Methods

Table 2 places the method among the baselines the original work reports. Its first eight rows are quoted from Table 1 of [1] under Sup-21K, averages over three seeds; the last two are our own runs at seed 42 through the code path of Sect. 4.1.

Table 2. Final average accuracy (%) under Sup-21K. Rows above the rule are quoted from Table 1 of [1], averaged over three seeds; the two below are ours at seed 42, from a different implementation and seed set. Bold indicates the best value within each group, not a controlled cross-group ranking. S-Prompt++ and HiDe-LoRA are the variants evaluated in [1].

Method	CIFAR-100	ImageNet-R	ImageNet-A	5-Datasets
DualPrompt [5]	86.60	68.79	39.76	76.09
S-Prompt++ [16]	88.81	69.68	39.10	83.38
CODA-Prompt [7]	86.94	70.03	30.96	62.54
LAE [17]	85.10	72.25	40.36	64.16
CPrompt [18]	83.18	69.24	37.60	71.72
InfLoRA [19]	85.28	72.24	42.12	71.97
HiDe-LoRA [3]	88.43	71.90	42.18	93.25
HRM-PET [1]	89.45	73.86	44.28	93.99
HRM-PET, our run	89.77	74.02	45.85	91.37
Proposed	90.53	75.51	47.42	93.77

Our seed-42 baseline differs from the published mean by +0.32, +0.16, +1.57 and -2.62 points on the four datasets, respectively. These reproduction gaps preclude a controlled comparison across the two groups. The controlled comparison remains the paired baseline–full difference in Table 1.

4.4 The Full Grid

The original work covers four datasets by five pre-training configurations; we rebuilt all twenty cells, evaluating both methods through one code path at seed 42. Every cell improves on both accuracy axes: Acc@task by +0.97 to +4.57 and Acc@1 by +0.59 to +3.58, against baselines spanning 31.8 to 93.8 Acc@1. Retention is descriptive rather than conclusive: final accuracy on earlier tasks rises in all twenty cells, by +1.83 on average, while accuracy at learning time rises by +1.58, so Backward, their difference, stays small. It worsens in five cells, which by that identity means learning-time accuracy rose more there — a widening gap, not a fall in final accuracy.

4.5 Ablation

We ablate *cumulatively*, as the original work does, on ImageNet-R at seed 42 across its five pre-training configurations, whose baselines run from 68.47 to 74.02 Acc@1. Only differences are reported, since only they carry the argument.

Routing gains depend on the gate. Each stage alone beats the baseline on every backbone — routing by +0.29 to +0.74, the classifier by +0.51 to +3.07. Adding routing to class fusion moves Acc@1 by -0.07 to $+0.02$ with the gate off and by $+0.13$ to $+0.77$ with it on, positive on all five backbones. The classifier’s own contribution is smallest by far on Sup-21K, $+0.51$ against $+2.23$ to $+3.07$ elsewhere — the backbone every design decision was taken on.

Conditional contributions over four seeds. On all three datasets we ran ungated class-only fusion ($w = 1$, $\beta = 0.5$, no gate), the same with the margin gate, and the full method, at seeds 42–45. Full minus gated class-only measures routing conditional on gated class fusion; gated minus ungated class-only measures the gate without routing fusion. These are conditional contrasts, not independent stage effects.

The conditional routing effect is $+0.083$ [$+0.026$, $+0.140$] on ImageNet-R, -0.030 [-0.085 , $+0.025$] on CIFAR-100 and $+0.156$ [$+0.055$, $+0.257$] on CUB-200. The gate is $+0.603$ and $+0.355$ on the first two, both excluding zero, and -0.284 [-0.570 , $+0.002$] on CUB-200. Each pair sums to the full-against-ungated-class-only change, $+0.69$, $+0.33$ and -0.13 , to within 0.005. Routing is positive on two datasets and unresolved on CIFAR. The gate’s CUB estimate is negative but its interval includes zero; this does not establish harm. Full still improves over HRM-PET in Table 1. Only the first fusion stage changes Acc@task.

4.6 Sensitivity to the Mixing Weights

The two weights were fixed sequentially, w before the second stage existed and β afterwards, so the second stage may have shifted the optimum of the first.

Table 3. Acc@1 over the $w \times \beta$ grid, ImageNet-R at seed 42, full method with the margin gate.

	$\beta = 0.3$	$\beta = 0.5$	$\beta = 0.7$
$w = 0.6$	75.28	75.61	75.29
$w = 0.7$ (reported)	75.22	75.51	75.24
$w = 0.8$	75.12	75.50	75.27

Table 3 reports Acc@1 over the joint grid on ImageNet-R at seed 42 with the margin gate active.

The grid favours $\beta = 0.5$ at every w . At that β , Acc@1 spans 0.11 points, versus about one point for Acc@task; aggregate scores cannot identify which routing repairs improve classification or establish equivalent weights. The maximum is $w = 0.6$, not the reported 0.7. Across four seeds, 0.6 gains +0.332 [+0.143, +0.522] Acc@task and +0.108 [−0.064, +0.280] Acc@1 over 0.7 with the gate and $\beta = 0.5$; the Acc@1 sign reverses on seed 43. Without class fusion ($\beta = 0$), its Acc@1 gain is +0.170 [+0.136, +0.204]. We retain the earlier routing-stage choice $w = 0.7$ for consistency, not as an established full-method optimum.

4.7 Limitations

Design choices were explored on ImageNet-R, and CUB-200 also carries development experiments, so neither is an untouched holdout; CIFAR-100 tests transfer, but more seeds do not remove selection bias. CUB can also overlap ImageNet pre-training images [25]; overlap was not audited here. The twenty-cell grid, five-backbone ablation and joint weight grid are single-seed; the three-dataset conditional contrasts use four seeds. The conditional routing effect remains unresolved on CIFAR. Post-fix verification is incomplete as noted in Sect. 4.1. We lack a same-protocol RanPAC comparison, a full log/config manifest for the historical grid, and cross-checkpoint verification of the fixed feature map. Quoted baselines use another implementation; our 5-Datasets result is below the published figure. We claim no consistent retention benefit or equivalence and have not measured phase-level costs or sample-level repair/harm counts.

5 Conclusion

A fixed-feature RP head complements HRM-PET’s task proposal and final classification, adding one adapter forward per image and one ridge fit per task. Reported Acc@1 gains are 0.69–1.50 points across three datasets. Routing’s contribution depends on class fusion and its gate; the negative CUB gate estimate is unresolved. Post-fix verification remains incomplete, and the accuracy gains do not establish reduced forgetting.

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